Question 5

a) X represents the number of correct answer a student gets on the exam

$X \sim B(6, \frac{1}{4})$

b)

x	0	1	2	3	4	5	6
$P(X=x)$	0.178	0.356	0.297	0.132	0.033	0.004	2.441×10^{-4}

$$P(X=x) = \binom{n}{x} \times p^{x} \times (1-p)^{n-x}$$

$p = \frac{1}{4}$

$n = 6$

$x = 0, 1, 2, 3, 4, 5, 6$

c)

mean, $\mu = np$

$= 6\left(\frac{1}{4}\right)$

$= 1.5$

d) $P(X \leq 3) = p(X=0) + p(X=1) + p(X=2) + p(X=3)$

$= 0.963$

e) $P(\text{first incorrect answer occurs in fourth question}) = \left(\frac{1}{4}\right)^3 \cdot \left(\frac{3}{4}\right)$

$= \frac{3}{256}$